

Chapter 5 — Authority and Delegation

Field	Value
Status	Architect draft complete; constitutional and engineering review complete
Version	0.1
Scope	Authority sources, policy evaluation, delegation enforcement, protected actions, and audit
Constitutional basis	Decisions 5, 9–10, 16, 21, 25, 27, 31–33, 35–36, 39, 46, 48–50, and 58
Owner review	None required at this stage

1. Purpose

Authority answers one question only: “Who is permitted to cause this effect, in this context, now?” This chapter defines a policy-governed authorization system that consumes identity, authentication, delegation, context, evidence, and transaction facts. Trust, capability, relationship, ownership, and authentication may inform evaluation, but none automatically grants permission.

2. Authority Sources and Precedence

Source	Meaning	Limit
Constitutional Owner	The sole holder of constitutional authority.	Protected actions still require policy and, where required, the Owner Authorization Ceremony.
Constitutional invariant	A non-ordinary constraint on what HAL may become or permit.	Cannot be bypassed by role, delegation, urgency, or provider capability.
Policy grant	A rule-defined authorization under declared conditions.	Must be current, deterministic, and auditable.
Delegation object	A bounded grant from an authorized delegator to a recipient.	Cannot exceed, outlive, or broaden its source authority.
Transaction authorization	A short-lived authorization to execute one validated effect.	Bound to exact scope, policy version, delegation chain, and commit rules.
Emergency procedure	A policy-defined exceptional path.	Never removes identity, audit, evidence, or post-incident accountability.

3. Authorization Decision Model

The Authority Service evaluates a structured Authorization Request. It returns Allow, Deny, Require Step-Up, Require Owner Ceremony, Require Verification, Defer, or Escalate. A decision is an immutable Policy Decision Record, not an undocumented boolean.

Required input	Purpose
Actor and session	Establish current identity and authentication assurance.
Requested effect	Names target, operation, resource, data classification, side effects, and blast radius.
Intent and transaction context	Links the effect to approved purpose, plan, correlation, and recovery contract.
Delegation chain	Proves source authority, scope, purpose, conditions, depth, expiry, and revocation state.
Policy and constitutional context	Pins policy version, invariant state, operating mode, Treaty, and applicable constraints.
Evidence and verification state	Supplies freshness, risk evidence, required tests, and uncertainty.
Environment and time	Supplies approved contextual facts; neither network location nor relationship substitutes for permission.

4. Evaluation Sequence

- Authenticate the actor and validate session freshness, assurance, and liveness requirements.
- Resolve the requested effect to a capability contract, data/resource scope, and risk class.
- Load applicable constitutional invariants and deterministic policy versions; reject an invalid or ambiguous policy set.
- Validate delegation from the originating authority through every link; enforce scope intersection, conditions, expiry, revocation, and depth.
- Evaluate treaty, privacy, environment, resource, transaction, verification, and operating-mode constraints.
- Produce a signed decision record with rationale, constraints, evidence references, expiry, and any required step-up or ceremony.
- For allowed execution, issue a narrowly scoped, short-lived, non-transferable execution envelope; re-evaluate at commit barriers.

5. Delegation Enforcement

Delegation is subtractive: each link can only retain or narrow authority received from the previous link. An execution envelope is never a general credential. It can authorize only the specified effect, subject, resource, purpose, time window, conditions, and transaction context.

Rule	Enforcement
No ambient authority	Secrets, service identity, hardware possession, or provider integration do not create permission.

Rule	Enforcement
Scope intersection	The effective authority is the narrowest scope across Owner/policy/delegation/transaction/Treaty constraints.
Purpose binding	A grant may be used only for its recorded purpose and approved intent relationship.
Fresh revocation	Invocation and commit validate current revocation state; cached envelopes do not survive revocation.
Attribution	Records retain originator, executors, every delegation link, policy version, and decision rationale.
No silent escalation	A missing grant, uncertain policy, or unavailable evaluator yields defer, deny, or safe restriction—not inferred access.

6. Protected Actions

Protected actions include constitutional amendments, protected policy changes, Owner authority or succession changes, identity recovery, quarantine release, capability-class activation, Treaty activation, protected deletion, and changes to trust/authentication requirements. These actions require the Constitutional Kernel, risk-scaled verification, and the Owner Authorization Ceremony when Book I requires it.

- Agreement in ordinary conversation permits proposal preparation but is never a protected execution credential.
- Authorization binds to the exact immutable change object. A material change invalidates prior authorization.
- The safest reversible default applies when evidence, identity, policy, time, or ceremony requirements are incomplete.
- Emergency urgency may change scheduling and notification, never the constitutional proof burden or immutable audit requirement.

7. Interfaces and Evidence

Interface	Result
Authorize effect	Signed Allow/Deny/Step-Up/Verify/Escalate decision with rationale and expiry.
Issue execution envelope	Short-lived scoped envelope bound to decision, actor, transaction, and capability contract.
Revoke delegation or envelope	Durable revocation event and immediately effective policy state.
Explain decision	Human-appropriate explanation plus forensic references to policy, authority, evidence, and constraints.
Audit query	Authorized read of decision chain, approvals, execution

Interface	Result
	results, and outcome evidence.

8. Guarantees

- HAL MUST authorize effects, not identities in the abstract.
- HAL MUST NOT derive authority from trust, relationship, proximity, network location, model confidence, or capability availability.
- Every protected action MUST have a deterministic policy basis, immutable decision record, and required verification and Owner authorization.
- Delegation MUST be explicit, narrowable, revocable, attributable, and bounded by time and conditions.
- Authorization decisions MUST be explainable and reproducible from pinned policy, evidence, and context.
- If authority cannot be established, HAL MUST safely defer, deny, or escalate.

9. Constitutional Traceability Audit

Decisions	Coverage
5, 9–10, 16, 21, 25	Single authority, evidence-based consequential decisions, capability constraints, orchestration, hard policy limits.
27, 31–36, 39	Authentication, delegated authority, Owner-defined intent, tiered change control, transactions, capabilities, governed secrets.
46, 48–50, 58	Intent stewardship, sole Owner, sovereignty/Treaties, verification, amendment and authorization ceremony support.

10. Internal Engineering Review

Check	Result
Constitutional conflicts	None. Authority remains independent from identity, trust, authentication, and capability.
Escalation safety	Explicit evaluation, envelope binding, revocation, and protected action controls are included.
Auditability	Every outcome carries pinned policy, evidence, delegation, and transaction references.
Owner review required	None. The chapter implements existing constitutional authority rules without changing Owner powers.

11. Completion Status

Chapter 5 is complete. Chapter 6 will specify Intent and Planning Architecture, which provides the purpose and plans that authorization evaluates without granting authority itself.